CHAPTER 2



EVOLUTION OF A NETWORK

The formation of NASA had an immediate, galvanizing effect on United States space activities. One major effect was bringing together organizations and programs that up until then had been functioning as separate entities. Consolidated along with the NACA were: Earth science satellite and lunar probe programs from the DOD's Advanced Research Projects Agency; the Army's JPL in Pasadena, California;[1] the Army Ballistic Missile Agency's Development Operations Division in Huntsville, Alabama including the important Saturn launch vehicle project; the IGY Vanguard and Explorer satellite programs, and with it, the Minitrack Network.[2] Within weeks, Minitrack shifted from military control to NASA management. As part of this reorganization, contracts to industry were expanded to allow contractors to begin assuming the day-to-day network operations that had previously been performed by the military.

One such contractor was Bendix. Murray T. Weingarten, President and Chairman of the Board for Bendix Field Engineering Corporation (BFEC) from 1973 to 1989, would recall that in this arrangement,

> The contractor forces were able to respond quickly, yet under control, to the many needs of NASA, whether they involved manning a far-off tracking station or chartering and flying a Douglas DC-3 aircraft to calibrate these tracking stations. . . . The word went out, if you had a problem, whether it involved calibrating a tracking station or a need for unique equipment, call 'Hunter 6-7700', which was Bendix's local phone number in Maryland. It was a fast and cost effective means to operate, a fact confirmed many years later through numerous congressional oversight hearings.[3]

NASA depended on a large contingent of contractors to operate its tracking networks. This government/contractor "marriage" was a crucial ingredient as to why its tracking and data network has been as successful as it has been. Weingarten would attribute this long-term success to an infectious, esprit de corps of those who made the network happen.

> Whatever needed to be done, we did in an efficient and cost effective manner. The spirit was there. We were there to assist NASA in helping the United States become the first nation to land a man on the Moon. In those early days, NASA had many talented, strong, dedicated people who were decision makers gathered from other government agencies. We in industry became part of that team and were able to participate in the planning and the execution of those plans. The program had top priority and the team concept prevailed. Of course, there were problems, but we did not allow them to fester. They were discussed openly and resolved on a timely basis and the program moved forward. The program was a pioneering effort, and new program techniques, such as award fee concepts, were developed, all helping to move the program forward effectively and efficiently. This enabled us to break new ground, establish new policies, practices and procedures to have effective operations in all parts of the world, including such exotic places as Kano, Nigeria; Zanzibar, and Madagascar.[4]

To see how this "can do" attitude shaped the network, one needs to go back and examine how things began to change soon after NASA was established.

Soon after the Agency's formation in the fall of 1958, many of the people directly responsible for the Minitrack Network transferred from the Office of Naval Research and the White Sands Missile Range to what would become the Goddard Space Flight Center (GSFC). Even before NASA officially opened, Congress authorized in August of 1958, $3.75 million for a

"Space Projects Center" to be located near the Capital. On 1 August, Senator J. Glenn Beall of Maryland announced that the new Center would be located in Greenbelt just outside of Washington, DC on land that was already owned by the Agriculture Department's Agricultural Research Center. Plans for the new Center were approved by NASA's first Administrator T. Keith Glennan in November. To accomplish this transfer with as little disruption as possible to Project Vanguard, an agreement was reached between NASA and the Navy that provided for the continued use of Naval Research Laboratory facilities until the new Greenbelt facility could be completed.

It was around this time that Wesley J. Bodin, a sonar researcher with NRL, became curious about the new space program, enough so that he transferred into the first group at the new Goddard Center. "When I say transferred, it was more than people," said Bodin. "We transferred people, equipment, and functions. We had a network already built with the Army Corps of Engineers [Minitrack]. . . . This operating network was transferred—completely—into NASA."[5]

Initial personnel transfer took place on 30 November with 157 members from the NRL being designated to what was temporarily the Vanguard Division of the yet unnamed space center. A month later another 46 persons were moved to form the core of the Space Sciences Division. Over



American rocket pioneer Robert H. Goddard in a 1932 photograph. (NASA Image GPN-2000-001336)

the next three months, 73 additional people transferred to various new divisions. Major construction in Greenbelt soon began in April 1959.[6]

On 1 May, NASA officially announced that the Center would be named the Goddard Space Flight Center (GSFC) in honor of American rocket pioneer Robert H. Goddard who on 16 March 1926 launched the world's first liquid propellant rocket. Construction of the new Center continued over the next 18 months. Although the Center bustled with activity as soon as NRL personnel were transferred, NASA waited until 16 March 1961—the 35th anniversary of Goddard's pioneering launch—to officially dedicate the Center. In the meantiEme, NASA found office space wherever possible in the area to house the growing contingent of engineers and scientists. To maximize work space, Jack Mengel's people in the Space Communications Branch, even turned barracks into offices at the nearby Naval Station in Anacostia.[7]

Organizational changes soon followed. From the two groups that came over to NASA—the Vanguard group and a Space Sciences group—Jack Mengel formed a new networks division while John Townsend headed up the sciences division. (Mengel served as the first Director for Tracking and Data Systems at GSFC until retiring in 1972. Townsend became the Goddard Center Director from 1986 until his retirement in 1990.) Mengel would recall that:

> The excitement of the space program became a virtual magnet which attracted the very best people to handle the many new functions which had been assigned to the Goddard Center. This, of course, included tracking, data processing, computer operations, and worldwide communications which were needed to tie the system all together. Fortunately, we were able to assemble a truly priceless team of very dedicated and talented people.[8]

Part of the new Center's early organization included the Space Task Group (STG), a cadre of engineers and scientists out of Langley Research Center who were responsible for planning NASA's "manned satellite" activities. Although located at Langley, the STG was actually part of Goddard's early responsibilities as it assumed the lead for NASA's new space projects. It was a busy time. The Center grew so rapidly that most of the people working human spaceflight—commonly known as manned spaceflight—also had to move off-Center for a while into a rented facility some five miles away because the Greenbelt Center literally ran out of space.[9] In its multirole function, the Center was assigned the broad scope of managing scientific, communications, and meteorological satellite projects, and developing sounding rocket and spacecraft experiments. To support these activities, NASA's first real-time computer complex was built at the Center. The building that housed this new computer facility was among the first to be constructed at Goddard. Some

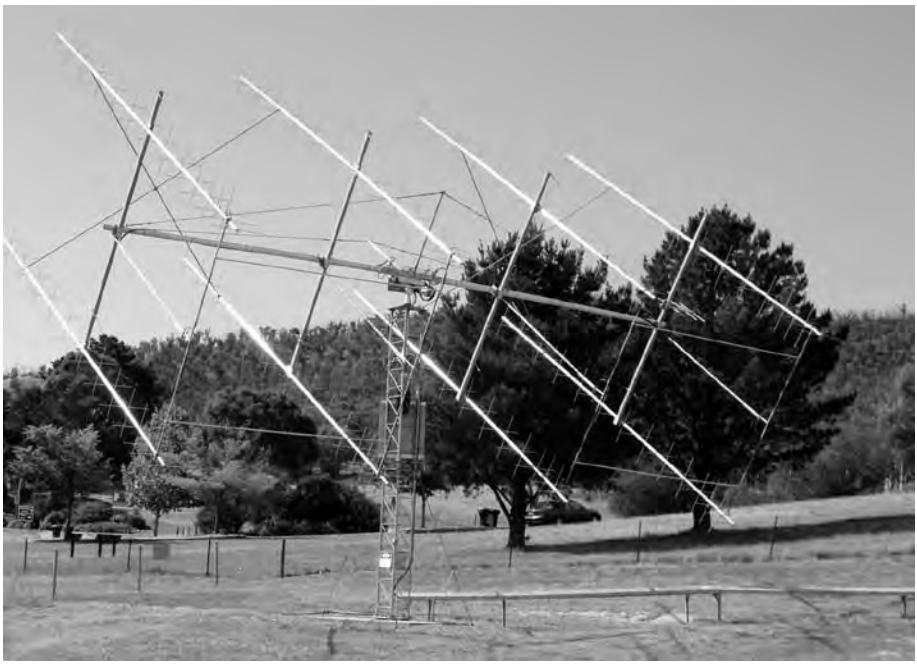


Dedication of the Goddard Space Flight Center in Greenbelt, Maryland on 16 March 1961. (NASA Image No. 022)

recall that it was still under construction and surrounded by a sea of mud even as IBM contractors installed interfaces to Cape Canaveral.[10]

In January 1961, four months before Alan B. Shepard, Jr. became the first American in space, some 670 people from the STG separated from the Goddard organization to form the nucleus of what went on to become the Manned Spacecraft Center (MSC) in Houston, Texas. In a move that would engender dissension between Houston and Greenbelt for years to come, Goddard still retained responsibility for managing and operating the tracking and data network for the human spaceflight program. It thus became the focal point from which all network activities (excluding the DSN which remained with the JPL) were coordinated and directed, and where data from the individual field stations were collected and processed.

While Minitrack continued to track Explorer and Vanguard satellites using radio interferometry, changes were being made during this time to accommodate technology advancements rapidly occurring in the field of RF transmission.[11] In 1960, Minitrack equipment was adjusted by network engineers to receive satellite transmissions in the Very High Frequency (VHF) part of the electromagnetic spectrum at 136 MHz. This modification greatly allevi-



A VHF double-stacked, yagi antenna, similar to common rooftop antennas used to receive over-the-air television signals, but much larger. The antenna consists of both vertical and horizontal dipoles which collect signals from a satellite, passing them to a preamplifier which boosts their strength for processing by a telemetry receiver. The photograph shows an antenna now at the Canberra Deep Space Communication Complex in Tidbinbilla, Australia, that is used to support the Interplanetary Monitoring Platform 8 (IMP-8) satellite. (Photograph by the author)

ated low frequency interference at many of the sites and was implemented by NASA so as to conform with new international standards being recommended by the International Telecommunication Union in the burgeoning field of space communications. In addition, telemetry reception at many network sites was improved with the addition of nine and sixteen yagi antenna arrays.

As launch vehicles became steadily more reliable in the early 1960s and satellites were being launched into higher inclination orbits (the tilt of a satellite's orbit measured with respect to the Equator), their ground tracks went well outside the ±35° latitude window that Minitrack had been designed to support. For the first time, polar orbits, where the ground track of a satellite covered the entire globe (ideal for weather observation and surveillance), came onto the scene. For this reason, Minitrack stations were soon added in

northerly and southerly locations outside the original "picket line," this even while some stations were being shut down. From 1959 to 1960, Minitrack reached a plateau in terms of the number of network stations.

By the early 1960s, a whole new generation of scientific and applications satellites boasting capabilities well beyond those of the IGY satellites sprang to the forefront. These ranged from weather, communication, surveillance, and Earth science spacecraft to lunar and planetary deep space probes. The orbit mechanics of these higher altitude satellites meant slower angular rates with respect to a given location on the ground as a satellite circled the globe. Elliptical orbits also meant that fewer stations could observe a satellite as it circled the globe. Some, like the Interplanetary Monitoring Platform (IMP), were on extremely high eccentricity orbits with apogees out to 240,000 kilometers (150,000 miles). Since interferometer effectiveness depended on measuring angular differences, the data that Minitrack could provide for precise orbit determination became increasingly inadequate. On top of that, modulation techniques also advanced to accommodate the ever increasing demand for higher communication data rates, more complex ground command operations, and higher bandwidth digital telemetry items such as television.

To accommodate this new generation of artificial satellites—soon to be followed by something even more ambitious in "*manned* satellites"—an evolution in NASA's tracking network had to take place. "While the Vanguard program was intended to plow new ground, show new rocket technology and develop new guidance and control systems," said Ches Looney when recalling the importance of these changes that were seemingly taking place overnight, "the manned space flight program, including its tracking system, required the use of proven, off-the-shelf hardware, to offer maximum reliability, because after all, human lives were [going to be] at stake."[12]

Looney, who was the Associate Division Chief for Advanced Development at Goddard from 1959 to 1966, added:

> We began to realize that while Minitrack was a marvelous system and absolutely essential in its time, the basic approach used was leading us into a blind alley. As things progressed, we were able to improve satellite trajectories and orbits as well as orbit calculations, which together with a new S-band ranging system [a higher frequency capable of handling higher data rates], allowed us to use large parabolic antenna dishes without independent acquisition aids such as Minitrack.[13]

In another sign of the changing times, computer telemetry processing facilities were installed at the field stations to directly feed into Goddard computers. This was a far cry from Minitrack, when tracking data was recorded

and had to be shipped to Goddard by air or surface courier, a process that took anywhere from two weeks to a month. Although coarse determination of the orbit track could be done at a station, precise orbit computations would be unknown for a couple of weeks.[14]

The genesis of these changes can be traced back to December 1959 when NASA formulated its first comprehensive program plan as the country's new space agency in a document simply entitled "NASA Long-Range Plan," prepared by its Office of Program Planning and Evaluation. The report that the Agency came up with was essentially a forecast of what it could do within certain budgetary constraints and reasonable extrapolations of the state-of-the-art in launch vehicles—the pacing facet of space technology at the time. Plans such as this were being modified continually, particularly in the face of Soviet space "firsts," which at the time seemed to be occurring at an alarming pace.

Satellite launches over the coming two to three years were forecasted as part of this plan. These projections provided Goddard engineers with a set of requirements from which specific approaches for comprehensive tracking, telemetry, and command (TT&C) design were selected, developed, tested, and eventually turned into operational hardware. One of the first items of business began in 1960 when the GSFC started a process to upgrade the TT&C equipment at existing Minitrack stations. In addition to modernizing these old Minitrack sites, new stations were established. The network that emerged became known in the space community as the Satellite Tracking And Data Acquisition Network, or STADAN.[15]

Over the next two years, large aperture, high gain circularly polarized, parabolic antennas replaced the less efficient, linearly polarized, yagi antennas so as to accommodate communications using the more bandwidth efficient S-band frequencies (2,100 MHz). The Minitrack approach gave way to far better equipped 12-meter (40-foot) and 26-meter (85-foot) dishes. The large 26-meter parabolic dish antennas, in particular, were designed to support the Nimbus meteorological satellite program with its flood of high data rate telemetry cloud cover photographs. In addition to receiving downlinked telemetry, these antennas could transmit satellite commands from a single disc-on-rod uplink antenna mounted on the side of the dish connected to a high power amplifier at the base of the antenna. At the same time, a somewhat scaled down, 12-meter version of the same antenna type was installed at a number of stations to circumvent the cost of the larger dishes.[16]

A 26-meter pointing antenna system was a pricey item, costing some $910,000 (in 1961) to design, engineer, fabricate, and erect. Mengel was to recall that in 1961, NASA Administrator Keith Glennan said never to ask him again for any more such pricey equipment.[17] However, this was only a harbinger of things to come as more advanced spacecraft and elaborate missions soon called for the use of even more sophisticated systems in the STADAN.

Just as critical to the STADAN mission of downlink telemetry acquisition and uplink satellite commanding was the tracking function. It had become a great deal more difficult as NASA's programs broadened to embrace satellites in high eccentricity, high apogee orbits all the way to geosynchronous orbits. In these orbits, angular rates with respect to a ground station changed very slowly (virtually none in the case of geosynchronous). Because of this, interferometry effectiveness was greatly reduced and sometimes eliminated altogether.

These tracking problems were anticipated by NASA early on. Addressing the issue, Edmund J. Habib at Goddard and Eli Baghadady of Adcom, Inc., devised a solution called the Goddard Range And Range Rate (GRARR) system. GRARR differed from Minitrack in that it was an *active* rather than a *passive* tracking system. In Minitrack, a beacon aboard a satellite continuously sent out a signal which was received on the ground when it came into range of a station. With the GRARR, a concept called "side-tone ranging" was used. In side-tone ranging, range tones were sent from the ground station up to the spacecraft on a VHF carrier separate from the S-band uplink carrying the command signal. An onboard transponder then returned these VHF tones along with the downlinked S-band telemetry. At the ground station, the precise time delay between the transmitted and received tones was measured to determine the range (and its rate of change) from the site to the spacecraft. With similar information from other STADAN stations from around the world, the orbit in which a satellite is traveling could thus be determined very precisely. When the system was first set up around the network, ranges out to 400,000 kilometers (250,000 miles or approximately the distance of the Moon from Earth) could be measured. Typical ranging accuracy was ±25 meters (82 feet), impressive for its time; range rate accuracy was just as good at ±15 centimeters (6 inches) per second.[18]

In describing the complexity of ranging systems, George Kronmiller, former Goddard GRARR Project Manager explained:

> During the early VHF years, our commanding and telemetry units were the same as the spacecraft's. We never took telemetry through the ranging system, not because we couldn't receive the signal, but because we didn't have the data handling gear to capture telemetry data and transmit it to NASCOM [NASA Communications Network] lines. Actually, until the STDN [Spaceflight Tracking and Data Network which Goddard formed in 1975], commanding and telemetry were kept separate from range and range rate, even with S-band. With the development of USB [Unified S-band], ranging and commanding could be handled on the same carrier. After the MSFN [Manned Space



Women operators, such as Melba Roy at the Goddard Space Flight Center in a picture taken in 1964, were a big part of the tracking workforce. (NASA Image GPN-2000-001647)

> Flight Network] and the STADAN combined to form the STDN, VHF remained the same, but the S-band system combined into a unified command, ranging and telemetry system using the same NASA standard near-Earth transponder on the spacecraft.[19]

GRARR equipment was usually housed in two trailers, one accommodating receiving and the other transmitting equipment. It operated in two different frequency bands: VHF (Very High Frequency) at around 150 MHz and an Ultra High Frequency (UHF) band at about 2,500 MHz. Two antennas were used, one for VHF and the other for UHF. Each was used for both transmission and reception. Angular tracking measurements could also be made by this equipment, but its accuracy was only good to within ±0.1 degree in elevation and azimuth, sufficient for the tracking of distant spacecraft but not always good enough for following spacecraft whose angular rates changed more rapidly. Thus, the angular information was often used as pointing or acquisition information for other instruments. Each of the two uplink transmitters radiated at 10,000 watts of power—about 5,000 times the power of an outdoor "walkie-talkie" hand radio. Since it continuously radiated while the equipment was on, station staff had to be cautious so as to avoid

exposure, particularly at the microwave frequencies. A system of flashing red warning lights could be found around a station so that people knew when the transmitters were on.[20]

By 1962, the initial GRARR ground elements were installed in trailers at Rosman, North Carolina; Tananarive, Madagascar; and Carnarvon, Wester Australia. The system was continually improved and used throughout the 1960s, supporting numerous Goddard satellite programs to provide range and range rate data whose accuracy would not be surpassed until the use of lasers a decade later. On Explorer 35—a so-called "Anchored Interplanetary Platform" orbiting the Moon—its range was measured to within 1,500 meters (4,900 feet) and the range rate accurate to within 65 centimeters (25.5 inches) per second at the lunar distance. Goddard engineers eventually considered the system reliable out to 1.3 million kilometers (800,000 miles), or three times the distance from Earth to the Moon. This was quite an accomplishment for 1960s technology.[21]

Yet a third improvement had to do with the dramatic increase in the data transfer volume that the Goddard network began to experience in the early 1960s. The observatory class of satellites such as the Infrared Space Observatory (ISO 1 in 1962) and the Orbiting Geophysical Observatory (OGO 1 in 1964) ushered in an era of information explosion. Meteorological satellites, like the Television Infrared Observatory Satellite (TIROS 1 in 1960) and Nimbus 1 in 1964, also required large amounts of telemetry to be downlinked to the ground during any given data pass over a ground station.

To meet these high "data dump" requirements, Satellite Automatic Tracking Antennas (known by the unflattering acronym SATAN) were installed throughout the network so as to supplement the telemetry reception workload of the parabolic dishes. These were essentially smaller, automatic tracking or "auto-track" replacements for the fixed Minitrack yagi antenna arrays. Operating in the VHF-band, SATANs were less sensitive (lower antenna gain) than their larger S-band counterparts, but were much less expensive to purchase, install and operate than the large dish antennas. Eventually, 30 SATAN antennas were installed throughout the STADAN during the 1960s.[22]

As the technology of NASA's global tracking network migrated from one of using radio interferometry and optics of the late 1950s to the more sophisticated STADAN of the 1960s, requirements on the locations of the stations changed with it. When the original Minitrack Network was developed, site selection was driven (and constrained) primarily by the short range of the tiny Vanguard transmitter, the technology of the receiving electronics and the general lack of precision tracking available during the satellite's early orbit. In other words, no one was exactly sure just where a satellite might appear over the horizon. Thus, Minitrack stations were originally spaced rather closely along a longitudinal line roughly following the 75th meridian (the "picket line"). Later, with

downrange tracking ships and much better ascent and orbit injection tracking, rough ephemerides (position-time information from which orbital elements are derived for orbit determination) were available even before booster burnout.

As a result, the three original "early ops" stations in the British West Indies—Antigua, Grand Bahama and Grand Turk—just downrange of Cape Canaveral were shut down in 1961. In some cases, actions by the DOD were a factor. A year earlier, for example, the Chula Vista station on Brown Field near San Diego had to be moved to Mojave when the Navy closed down that air station.[23] (It was at this new location that the Mojave Station went on to become the STADAN side of NASA's Goldstone Communication Complex.)

The picket line also changed. Take the Minitrack station in Chile for example. It had been located since 1957 at a spot called Salar del Carmen in the Pacific port town of Antofagasta on the northern part of the country. Initially operated by the U.S. Army, it was operated by a joint NASA/University of Chile team under contract NAS5-1925 when the space agency was established in 1958. However, with the improved capabilities of the nearby stations at Santiago and Lima, Antofagasta became redundant and was closed in July 1963. Some University of Chile personnel were transferred to the improved Santiago site (actually at nearby Peldehue) while most of the American contractors returned to the United States for reassignment. Telemetry equipment was redistributed, largely to the stations at Santiago and Lima. However, heavy equipment like power generators and air conditioners were returned to the U.S. for use at other stations.

While the Antofagasta site was being shut down, the neighboring station at Santiago underwent a $1.2 million improvement to include the installation of a 12-meter (40-foot) medium gain antenna, a 370-square meter (4,000-square foot) operations building and a new collimation tower (a tower located a few miles from the main antenna equipped with an RF emitter used as an aim point to checkout and calibrate the automatic tracking capability of the antenna). Under the original contract with the University of Chile which allowed the United States to put a station on Chilean soil, the school had "agreed to provide . . . land as may be necessary for the effective operation of the station."[24] Since the entire reservation on which the station was located was used only for pasteurization, obtaining the additional 10 acres that was needed for modernizing the station did not present a problem.

The station staff at Santiago almost immediately doubled in 1963, from 38 (16 Americans and 22 Chileans) to 62. Under the contract with the university, NASA eventually replaced half of the additional 24 people with trained Chileans. Said Wes J. Bodin, the Associate Chief of the STADAN Engineering Division at the time, this was just an example of the rapid growth that the STADAN saw during this time. "When a requirement came in, we

added to the network." Permanent or temporary sites were added as mission requirements called for. Bodin cited a couple of examples:

> When NASA started launching spacecraft into polar orbits, the second injection burn . . . was too far east of Johannesburg, so we added Tananarive. When we needed stations with large high-gain antennas, and this was a new requirement, we installed our first 26-meter (85-foot) antenna in Alaska. It wasn't too bad working there, so long as you were finished construction work by December when the temperature could hit 60 below![25]

This time period also saw what would be the first of several episodes in the political environment of the host country that would directly affect NASA's operation of its stations. Up until the 1950s, Cuba was a relatively stable independent island, enjoying strong agricultural trade with the United States and serving as a popular tourist destination in the western Caribbean. The U.S. had occupied this largest of the Caribbean islands in the Spanish-American War of 1898 until its independence was granted in 1902. After that, the United States continued to have a major influence in Cuban affairs, even occupying it briefly a second time from 1905 to 1909.

In 1940, Fulgencio Batista was elected president and over the next four years, began a series of idealistic reforms. Not surprisingly, the Cubans did not see eye-to-eye with his isolationist views and voted him out of office in 1944. In 1952 however, Batista regained power in a bloodless coup three months before the planned election. This time, he instituted a dictatorship. As a result, many factions and guerrilla groups began opposing him. One such group was led by Fidel Castro Ruz who had participated in a failed attempt to overthrow the Dominican Republic government in 1947. He was a staunch nationalist known for his opposition of American influence in Cuba. In 1953, Castro attacked the Moncada barracks, the main provincial garrison of Batista's army, but was captured and exiled to Mexico. However, he returned to Cuba in November of 1956 and over the course of the next two years, his "26th of July Movement" gathered strength and influence, eventually to the point where Batista finally had to flee the country on 31 December 1958. The next day, Castro established a one party communist state in Cuba, the first of its kind in the Western Hemisphere.

At the time of Batista's ouster, 75 percent of Cuba's farmable land was owned by foreign interest (mostly U.S.). The new government adopted land reforms and confiscated all the private property owned by upper class Cubans and foreign companies. Although Castro did not officially reveal his Marxist leanings until 1961, the United States had recognized the intentions of the new government, and as a result, relations rapidly deteriorated. On the

day that Castro seized control of the Cuban government, the Minitrack station, which was located at Batista Military Airfield, was shut down even as the American Embassy scrambled to learn the fate of NASA Station Director Chester 'Chet' Matthes and his family, as well as that of the military personnel assigned to the tracking station.[26] By noon the next day, confirmation was received that all were safe. Station personnel were then told by the State Department to follow instructions from the American Embassy and that the station would reopen when the Embassy received word and considered it advisable. But that word never came. Instead, they were told to close out the station, remove the assets and leave the country.[27]

In events unrelated to Cuba during this time, operations at the Fort Stewart Station in Georgia underwent some reorganization. When the Minitrack station was established there in 1957, the site was primarily an Army anti-aircraft training facility. After the Korean cease fire, the Army recognized that it needed to maintain a ready and able armor force that could be deployed at a moment's notice to deal with threats that may suddenly erupt. As a result, Fort Stewart was turned into an Armor and Artillery Firing Center since its old anti-aircraft ranges and impact areas were better suited for this purpose than for the new age of missiles. Because of this realignment, the NASA tracking station no longer fit the bill for the post's new mission and the Fort Stewart Station was phased out of the Minitrack Network.[28]

With the abrupt shut down of two primary stations, a gaping hole materialized in the middle of the picket line. NASA had to quickly find a replacement. Attention focused on Fort Myers, Florida. Situated essentially midway between Fort Stewart and Havana, it was a logical choice, geographically. Equipment was immediately relocated to Florida from the two sites that had shut down and satellite pass responsibilities redesignated and combined at the new site.

From 1960 through 1963, most of the early Minitrack stations were refurbished and transitioned into the STADAN. For example, Johannesburg, Quito and Santiago received 12-meter (40-foot) antennas, and all acquired dual receive/command SATANs. Santiago was modified into a GRARR site, as well. The South American stations, in particular, continued as a centerpiece in the evolved network, providing NASA with a tracking and data acquisition capability in the southern hemisphere.[29]

With satellite ground tracks now extending well beyond the original $\pm 35^{\circ}$ latitude window, new stations at locations outside that latitude range had to be built. From 1960 to 1966, the GSFC oversaw the build up of eight additional STADAN stations on four continents. These were the most advanced stations to date. In North America, they were located at Fairbanks, Alaska; East Grand Forks, Minnesota; Rosman, North Carolina; and St. John's, Newfoundland, the eastern-most point of Canada.[30] In Europe, the Winkfield Station was erected in Berkshire, England. The addition of

these sites greatly expanded satellite coverage in the northern hemisphere, increasing the observation window from 35° latitude to 65°. In the south, the Johannesburg Station was joined by a new one at Tananarive in the Malagasy Republic on the island of Madagascar.

Fairbanks and Rosman soon became the most extensive stations in the system. In March 1962, Gilmore Creek, on the outskirts of Fairbanks, received the first of the large parabolic dishes, featuring both the 12- and 26-meter (40-and 85-foot) dishes, along with the GRARR and a full complement of SATAN equipment. In November 1966, in what was essentially a cost saving move, the STADAN site in College, Alaska, still operating Minitrack equipment, was consolidated with the nearby Fairbanks Data Acquisition Facility (DAF) at Gilmore Creek for an estimated annual savings of $292,000.[31] This was a relatively easy decision by Goddard as these sites in the vicinity of Fairbanks were less than 40 kilometers (25 miles) from each other. With this move, the last of the remaining Minitrack antenna arrays was transferred to Gilmore.

The Fairbanks Station—also known officially as "Alaska"—was located in the central part of the state 22 kilometers northeast of Fairbanks at a well-known landmark where Rose Creek joined Gilmore Creek. Being the most northerly of all the stations, Alaska provided crucial support for polar and high eccentricity, elliptical orbiting spacecraft such as the Nimbus weather satellites and the Alouette. (On 29 September 1962, Canada became the third country to orbit an artificial satellite with the launch of the Alouette, a science satellite to study the affect of aurora borealis, or the Northern Lights, on radio frequency (RF) signal propagation in the upper ionosphere and its effect on communications.) Conditions at the station were at times harsh, characterized by cold winters of below -60°F and summers in which the temperature could rise above 90°F. During the winter months for example, the station had to operate a steam boiler system to forestall the spread of glacier formations around the site.

Even though work conditions for personnel assigned to remote locations such as Alaska could at times be trying, severe injury or death in the field was not a common thing. A case in point, though, was recorded on 3 May 1969 at the Fairbanks Station. At approximately 4:30 p.m., a RCA Service Company (the station's maintenance and operations contractor) employee named George Matilla was found crushed to death beneath his car at the station's garage. Since it was a weekend, no one knew exactly what had happened. His death was later ruled as an accident by the Alaska State Patrol who concluded that the jack had slipped as he worked alone under his car, fatally crushing him.[32] Throughout the years, there have been other stories as well, particularly at the more desolate locations, of those who wandered out-



Orroral Valley (ORR), nestled some 50 kilometers southwest of Canberra in the Namadgi National Park, was one of 10 NASA stations in Australia at one time. The most comprehensive Satellite Tracking And Data Acquisition Network station in the Southern Hemisphere, ORR operated from 1965 to 1984. Clearly seen are the 26-meter antenna adjacent to the main Operations Building and the 2 SATAN antennas. Note the Minitrack array in the foreground. (Photo courtesy of CSIRO)

side the station compound to check on equipment or simply "went fishing" and were never heard from again.[33]

In the Southern Hemisphere, cooperation between Australia and the United States in the field of space research began in 1957 with construction of facilities at Woomera, South Australia, for the IGY. In a formal exchange of notes in February 1960, the two governments agreed on a cooperative space program:

> The object of such further and extended cooperation would be to facilitate spaceflight operations contributing to the advancement of our mutual scientific knowledge of man's spatial environment and its effects; the application of this knowledge to the direct benefit of man; and the development of space vehicles of advanced capabilities, including manned space vehicles.[34]

The Australian ground stations were established and run by the Department of Supply, or the DOS. In this agreement, construction and

operation of the stations were financed by NASA while the DOS provided management and staffing, either from its own resources within the Weapons Research Establishment or through Australian industry contractors. In this way, Australia had the most autonomy and independence in running the NASA stations outside of the United States. Unlike most stations where the local staff reported to an onsite, NASA-appointed Station Director, the DOS in Australia appointed its own people. In this way, a great sense of ownership was promoted—one which prevails to this day.

The Australian government welcomed space activities as intrinsically good and providing a medium for strengthening diplomatic cooperation with the U.S. It viewed such activities as having strong spin-off potential to Australia in such fields as defense, communications, and astronomy. From the beginning, Australia was sensitive to the importance of retaining independence and having a substantial share in the control of network activities on its soil. Robert A. Leslie, who headed Australia's network operations—a "father figure" in the commonwealth's space work—stated its position very clearly in a 1974 letter to the Minister of Science in which he reiterated this right of Australian ownership:

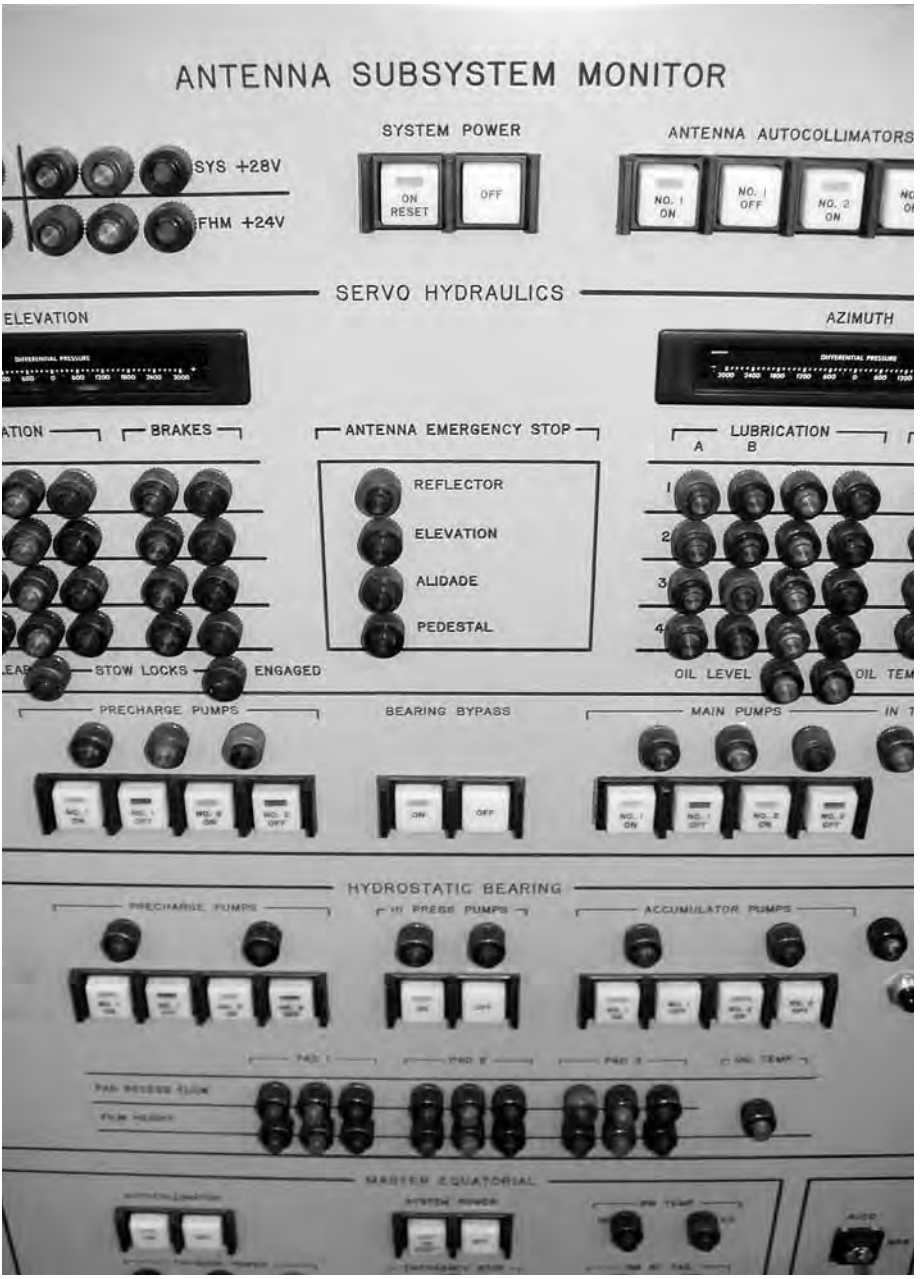
1 The employment of foreigners where Australians could be employed would be undesirable.

2 Control of the activity in foreign hands might have been a matter for question on security grounds and would have been a matter of national controversy.

3 The maximum spin-off to Australia—for example, to industry, universities and government agencies—could best be affected by Australians.

In return for what could be viewed as granting of Australian autonomy, the DOS made an annual contribution to NASA, a subsidy of $140,000 for operations of the Australian Tracking Stations. Although insignificant in terms of absolute dollar amount to NASA's budget (measured in the billions of dollars), this token sum has nevertheless been viewed importantly by the U.S. as a sign of goodwill and good faith on the part of the Australian government.[35]

In August 1960, the Minitrack station at Satellite Tracking Centre just northwest of Woomera was moved to the nearby Island Lagoon dry lake bed 22 kilometers (14 miles) to the south and its operations combined with the new Deep Space Network (DSN) site there. A station was also opened at Carnarvon, Western Australia, collocated with a Mercury Space Flight Network station that was then being constructed. However, the largest STADAN site in



A vintage 1960s Antenna Control Unit (ACU). Graphical computer displays, touch screens and joysticks have, for the most part, replaced the lights and buttons of the old ACUs. Shown is a control unit on display at the Space Museum at the Canberra Deep Space Communication Complex in Tidbinbilla. (Photograph by the author)

the Southern Hemisphere officially opened on 24 February 1966 at Orroral Valley 50 kilometers (30 miles) southwest of Canberra.

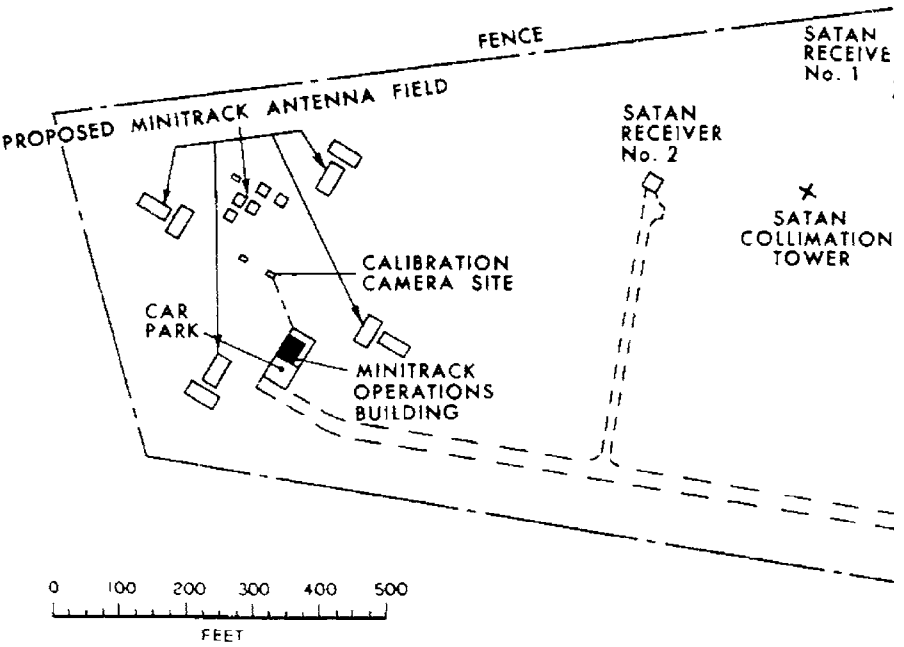
The Orroral Station was needed as Woomera (officially closed on 22 December 1972) was no longer a cost-effective location for NASA. Situated in the quiet valley of Namadgi National Park, Orroral provided an RF-quiet area to meet requirements for the new 26-meter (85-foot) antennas that were being installed across the STADAN. The rolling hills surrounding the area made Orroral an ideal environment for a tracking station by naturally shielding the site from man-made radio frequency interference. When the station site was chosen, the isolated valley located deep in the "Land of the Echidna and Platypus" was accessible only to light and off-road vehicles. Even today, one needs to cross the 100-year-old Tharwa Bridge just to reach the site. Before construction could start, suitable roads had to be paved for the passage of heavy construction equipment and antenna hardware. As at sites around the world, NASA provided this as part of its construction.

Six months after the station became operational, network responsibilities at Island Lagoon were phased out and what was left of the Minitrack equipment was moved to the new location. The net effect of this growth was to more than double the time that high inclination orbiting satellites could be in contact with the ground. Orroral became the most comprehensive STADAN site in the Southern Hemisphere. Construction of the station began in November 1964 and was completed in July of the next year at a cost of £1,000,000 (about $1.5 million U.S. dollars in 1964).[36]

The layout of the station was fairly typical of the new, larger DAFs across the STADAN. Equipment was contained within an area of about 40 acres and laid out so as to minimize interference to the receiving systems from buildings, support structures and other antennas. The centerpiece of the station was the 26-meter diameter parabolic tracking antenna. Weighing with its support structure and hydraulic gimbal-drive mechanism some 400 metric tons (882,000 pounds) and standing 36 meters (120 feet) high, it was easily the most prominent structure on site. From the Operations Building, the antennas were controlled and all tracking activities coordinated with the rest of the network.

During acquisition, the antenna—directed by data passed from Goddard—searched a specified area of the sky until it picked up the beacon signal transmitted by the satellite. The antenna data system then measured and encoded the look-angles and fed this information into a servo control system and an Antenna Control Unit (ACU) in the control center at the Operations Building. The information was also recorded on teletype punched tape and passed to the control center for orbit determination.

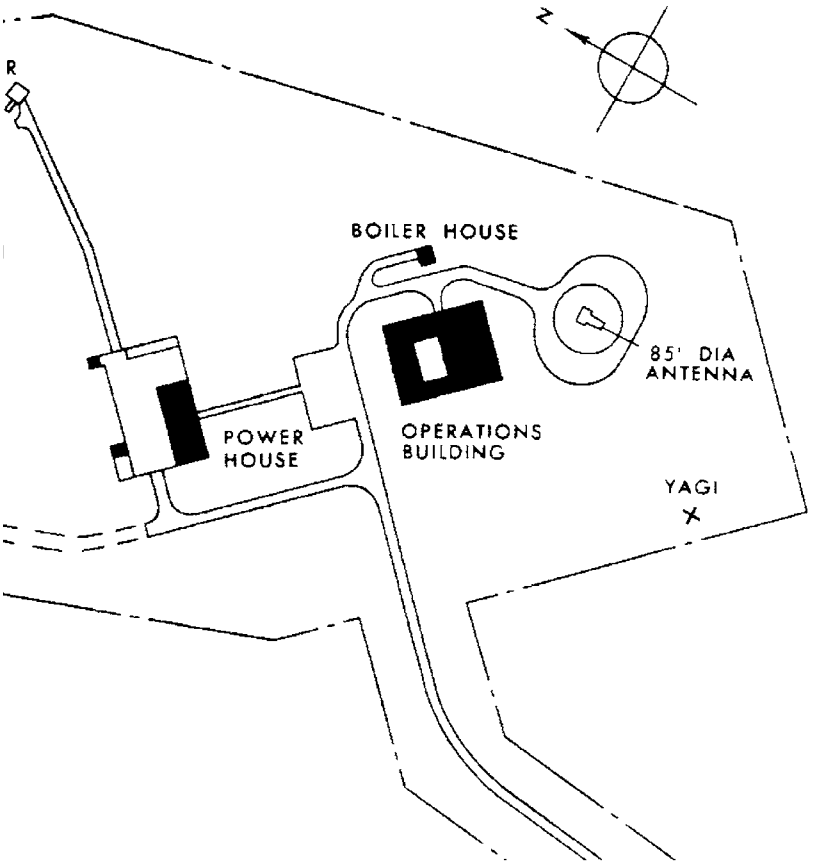
To receive telemetry downlink from a satellite, a command signal was encoded and transmitted using one of two 2,500 watt transmitters feed-



Layout of the Orroral Valley STADAN Station. Note that the Minitrack array transferred from Woomera was eventually placed further west of the SATAN receivers than shown

ing a small command antenna that was attached to the rim of the main dish. Signal from this uplink activated a transmitter in the satellite and data stored onboard the satellite could be transmitted to the ground station. Data and timing signals were then processed for recording onto 14-track reel-to-reel magnetic tapes and, in the case of time-critical information, for transmission by teletype back to Goddard. Backup receiving was usually provided by an onsite steerable Yagi antenna which could be directed by information from any of the primary tracking dish antennas.[37]

A continuous electrical power supply to a station was obviously vital to its operations. Even a brief outage could cause loss of data or compromise the mission if it were to occur at a key moment. If available, network stations used commercial line power backed up by its own diesel power gen-



in the diagram. (STADAN Facility, Orroral Valley, ACT: Information Brochure. Folder 680/5/23 NASA Australian Operations Office, Yarralumla, ACT)

erators. If line power was not available—as was the case at ORR—then power was generated onsite exclusively using dual-redundant generators with sufficient capacity to provide full backup should one unit fail. At Orroral, for instance, two 250,000-watt and two 500,000-watt units were used. Sites usually had two separate power distribution systems, one to run the station core electronics and tracking equipment and the other for utilities, air conditioning, lighting, and facility electricity needs.[38]

Finally, a DAF was not cheap to operate, requiring a staff of about 150 technical, administrative and maintenance workers rotating in shifts. The operating cost was about $1.2 million a year in the mid-1960s, almost as much as the onetime cost to build a station. NASA spent over $140 million in Australia alone in the 1970s. Besides just getting the technical job done,

such international cooperation also yielded social benefits. For example, by the mid-1970s, seven of the ten stations which had been built in Australia since 1958 had become popular tourist attractions, often offered as stops by touring companies. The Department of Science officially estimated that from 1965 to when it closed in 1974, the Carnarvon Station was visited by 7,000 to 10,000 people every year (this, in a continent of around 12 million in 1970). Similarly, the DSN site at Tidbinbilla on the outskirts of Canberra was drawing some 8,000 visitors annually.[39]

In addition to the permanent sites at Orroral and Carnarvon, several project-specific (temporary) stations were also added by Goddard in Australia. One reason that the Southern Hemisphere played host to most of the temporary sites was because that was where the critical Earth-injection phase of a mission to send satellites into high-inclination orbits usually took place. Key among these places was Darwin, a transportable station on the northern shores of Australia, selected specifically to support the Orbiting Geophysical Project missions, and Cooby Creek near Toowoomba in Queensland, which was added to support the Applications Technology Satellites (ATS). Kano, Nigeria in 1965 provided another temporary site for STADAN, briefly housing equipment supporting the International Satellite for Ionospheric Studies (ISIS). Goddard even ventured into Pakistan where a site was set up for a while to observe LANDSAT.[40]

As STADAN grew, one of its requirements was to receive high bit rate digital telemetry from a new observatory-class of large, science satellites. These included the Orbiting Geophysical Observatory (OGO), Orbiting Astronomical Observatory (OAO), Orbiting Solar Observatory (OSO) and the Nimbus weather satellites. While the facility in Alaska could support polar orbiting and high inclination satellites, it could not support this new class of observatories, which orbited Earth at lower inclinations.

For these missions, Goddard needed a wideband DAF at a latitude that was suitable for a high percentage of data recovery on these orbits. Dubbed internally as Project 3379, network engineers began looking for a suitable location for their newest station in 1962. Site surveys eventually whittled the number of possibilities down to two: Fort Valley, Georgia and Rosman, North Carolina. Rosman—the actual station site was some 15 kilometers northwest of the town itself—was eventually selected. The choice was due in part to Rosman being closer to Greenbelt which expedited experimenters with evaluation and control of their projects. Located 55 kilometers southwest of Asheville on the grounds of Pisgah National Forest, the site had an advantage of the natural setting of the surrounding hills which provided shielding from electromagnetic interference. Because it was on national forest land, the location also lacked high voltage transmission lines and commercial circuits in its vicinity



Natural RF shielding from the surrounding hills is evident, as is the size difference between a 64- and 26-meter antenna in this undated photograph of the DSN station at Tidbinbilla. (Unnumbered photograph, Folder 8820, NASA Historical Reference Collection, NASA History Division, NASA Headquarters, Washington, DC)

and, in the early 1960s, had an absence of commercial airline routes directly over the station.[41]

NASA invested $5 million in 1962 to build the station, filling a critical need as the network modernized from Minitrack to STADAN.[42] Rosman was established specifically as a full service Data Acquisition Facility and was among the first not to have the old interferometry tracking system. The site was well equipped to handle the new network mission, providing a full suite of the most up-to-date equipment for its time: telemetry reception with two 26-meter antennas that could autotrack at 1.7 and 2.29 GHz; a 3 kW command uplink system; a SATAN 16-element automatic yagi antenna array; S-band GRARR; a 1,200-square meter (13,000-square foot) operations building; a 420-square meter (4,500-square foot) power and service building housing four 200-kilovolt diesel generators, garage, and utilities; a 140-square meter (1,500-square foot) building to house the antenna hydraulic drive system; and a collimation tower with a small transmitter building located 2 kilometers (1.3 miles) west of the main antenna to serve as a boresight (beam center) for calibration and testing of the main tracking antennas. The first 26-meter system became operational in

July 1962 followed by the second unit in August 1964. These were immediately used to support S-band communications, which began earlier that year with the launch of the Hughes-built Syncom 1, the world's first geosynchronous communications satellite, on 14 February 1963.[43]

The Tar Heel state embraced its new scientific venture enthusiastically. On 26 October 1963, Rosman was officially dedicated with a ceremony replete with VIPs. Principal speakers included Governor J. Terry Sanford; U.S. Senators Sam J. Ervin, Jr. and B. Everett Jordan; Dr. George L. Simpson, Jr., NASA Assistant Administrator for Technology Utilization and Policy Planning; Edmond C. Buckley, Director of the Office of Tracking and Data Acquisition; and Dr. Harry J. Goett, GSFC Center Director. U.S. Representative Roy A. Taylor served as master of ceremonies.[44]

Although small compared to the 70-meter (230-foot) antennas used at the time by the DSN, the 26-meter (85-foot) dishes used in STADAN were nevertheless massive in their own right. Their movement was controlled by two large hydraulic motors that enabled three kinds of operation:

1 Manual operation in which the antenna was guided by an operator, who sitting at his control console, actually steered the antenna by sending electrical servo-signals to the motors.

2 Computer operation in which commands were sent by a computer that predicted the path of the satellite. Due to its large aperture, the 26-meter (85-foot) antenna had a rather narrow beamwidth (the direction in front of an antenna in which RF signals can be reasonably detected and focused). Consequently, it must be pointed near the horizon in the vicinity in which the satellite was expected to rise before the signal could be acquired (the acquisition process). By predicting the path of a satellite in orbit, the antenna could be pre-positioned by the computer thereby reducing the search time.

3 Automatic operation, or autotrack, in which a satellite's movement across the sky was automatically tracked by the antenna as it moved across its field-of-view. In this mode, the ACU converted the position of the satellite relative to the antenna boresight into electrical control signals which were sent to the antenna. In autotrack, the antenna was usually pointed just above the horizon at an azimuth where the satellite was expected to break horizon. Once acquired, autotracking enabled the dish to follow a spacecraft from horizon to horizon. During this process, the station computer converted the angular position of the

> antenna into electronic code which was automatically punched on teletype tape. This data was then transmitted to Goddard via the NASCOM where it was processed along with pass data from other stations for use in orbit determination calculations.[45]

Output from each of these antennas was fed to a telemetry receiver located in a nearby control and operations building. The job of the receivers was to convert the radio frequency signal from the antenna into a lower frequency signal which could then be recorded onto reel-to-reel magnetic tape—a process known as down converting. The output of the receiver contained many individual pieces of information, not unlike, for example, the picture and sound for television received on a single channel. The magnetic tape recorders used to record the telemetry operated on the same principles as everyday-use home tape recorders but were much more robust in terms of the amount of data and speed they could record at. In addition to the telemetry itself, a precise time reference, the signal strength and operator voice annotations, if any, were all recorded simultaneously onto the tapes. Once packaged, the tapes were shipped to Goddard for processing.

In spaceflight communications, precise timing data, in particular, is of the essence. A spacecraft in circular orbit at 480 kilometers (300 miles) above the surface of Earth travels at a speed of 7,600 meters-per-second (25,000 feet-per-second), or roughly 27,300 kilometers-per-hour (17,000 miles-per-hour)! Therefore, accurate timing information is of the utmost importance so that computation of the spacecraft's position in space can be made and that timing of various scientific events as downlinked to the ground can be pinpointed. In the STADAN, DAF timing was synchronized to the U.S. Bureau of Standards Time Standard Radio Station—WWV in Boulder, Colorado—to an accuracy of ± 0.001 seconds, or one one-thousandths of a second. While this level of accuracy may not seem like much in the age of the Global Positioning System, where timing accuracies are measured not in terms of milliseconds but *micro*seconds (one millionth of a second), it was quite the accomplishment in the analog era of the 1960s. These time-code generators produced electronically coded pulses which were recorded onto magnetic tapes simultaneously with the tracking and telemetry signals to provide the needed timing reference. Finally, to compliment the new, large, parabolic antennas, Rosman, along with the other DAFs at Gilmore Creek and Orroral Valley, continued to use the SATANs. Even though older, they still provided the much larger field-of-view needed to track the older generation satellites that still operated on VHF frequencies.[46]

During this time, the STADAN also saw the addition of one of the few network additions not in the English-speaking world. In 1964, a transportable station was set up on the island of Madagascar just off the east

African coast. A common language was obviously desirable from the standpoint of working with local nationals and with the local government. This was always one of the factors NASA considered when choosing locations for its ground stations. However, a station was required in the western part of the Indian Ocean for tracking highly elliptical, high apogee orbit injections that occurred over that part of the world. Also, NASA had to look for another location when it became apparent that the South African station would not be able to cover some of these critical events.

On 19 December 1963, the U.S. entered into a 10-year agreement with the Malagasy Republic allowing for the installation of a transportable ground station outside the port city of Majunga in northwest Madagascar. This agreement was reached in accordance with the spirit of a United Nations resolution calling for the application of results of space research to benefit all peoples. In addition to benefiting that region of the world by generating much-needed weather forecasts (especially during hurricane season), the station provided jobs for some 200 local residents in nontechnical positions for handling of day-to-day station maintenance. In reaching this agreement, NASA sent a delegation to the capital city of Tananarive where they were "received by the president, Mr. Philibert Tsiranana, most graciously in an office decorated with space memorabilia."[47] He soon gave the United States his enthusiastic support and permitted NASA to start bringing telemetry vans into Majunga.

The initial equipment consisted of five 9-meter (30-foot) trailers, each housing a 136.2-KHz and 400-MHz telemetry receiver that were geared to support the Nimbus, POGO (Polar Orbiting Geophysical Observatory) and A-12 Goddard satellite programs. Much of the equipment at Majunga came from the Australian sites of Muchea and Woomera, which had phased out at the conclusion of Project Mercury the previous May.[48] An MPS-26 radar was also temporarily deployed to support the EGO (Eccentric Geophysical Observatories) satellites.[49]

Explanatory literature handed out to familiarize station workers assigned to Madagascar described the environment as an area of mild winters and rainy summers, a relatively expensive but charming place to live. The handbook noted that the people of Madagascar were not politically minded and were predisposed to favor America and Americans. Harry McKeehan, who represented GSFC in negotiations with the Republic, called "our friendship with the president and the people of this island republic invaluable in building and operating this Indian Ocean site."[50] This cooperation was to play a pivotal role later when a political uprising in nearby Zanzibar created a tense situation, one in which American lives were put in jeopardy that required an evacuation to the Malagasy Republic (see Chapter 4).

Table 2-1: Stations of the Satellite Tracking And Data Acquisition Network (STADAN)

Station	Latitude Longitude	From	To	Equipment and Capabilities	Remarks
North America					
Alaska (Fairbanks)	64°59'N 147°31'W	1962	1984	GRARR, MOTS, Minitrack, SATAN, One each 40, 45, 85-foot dish	College site closed in 1966 and Minitrack equipment transferred to Gilmore. The most comprehensive of the STADAN stations.
Blossom Point, Maryland	38°25'N 77°06'W	1956	1966	Minitrack, MOTS, SATAN	The prototype Minitrack station. Used as a servicing station and training facility. NASA ceased joint operations with the Navy in 1966. Continued to be used by the NRL as a satellite control center.
East Grand Forks, Minnesota	47°56'N 97°01'W	1960	1966	Minitrack, MOTS, SATAN	Located near Grand Forks Air Force Base, a key ICBM station.
Fort Myers, Florida	26°33'N 81°52'W	1959	1972	Minitrack, SATAN, MOTS	Station formed when Havana and Fort Stewart shut down. Minitrack equipment transferred.
Mojave, California	35°20'N 116°54'W	1960	1969	40-foot dish, SATAN, Minitrack, MOTS	Moved from Brown Field near San Diego in 1960. Located in the Goldstone complex in valley adjacent to DSN site.
Network Test and Training Facility, GSFC, Greenbelt, Maryland	38°59'N 76°51'W	1966	1986	Minitrack, 59-foot antenna used for IUE support transferred to the Naval Academy. 30-foot antenna moved to Wallops in 1986	Until 1974, the NTTF was used to test equipment bound for the network and for personnel training. In 1974, it became part of the operational STDN. Station responsibilities were transferred to Wallops in 1986.
Rosman, North Carolina	35°12'N 82°52'W	1963	1981	Two 85-foot dishes, Three SATANs, GRARR, ATS	The most extensive STADAN site along with Alaska. Designed to receive high bit rate telemetry from observatory satellites. Turned over to the DOD in 1981.
St. John's, Newfoundland	47°44'N 52°43'W	1960	1970	Minitrack, MOTS	First of the northern latitude stations to support tracking of polar orbiting satellites

continued on the next page

Station	Latitude Longitude	From	To	Equipment and Capabilities	Remarks
South Amer ca					
Lima, Peru	11°47'S 77°09'W	1957	1969	Minitrack, MOTS, SATAN	One of the original Minitrack stations. Turned over to their university under contract.
Quito, Ecuador	00° 37'S 78°35'W	1957	1981	Minitrack, MOTS, SATAN, 40-foot dish	Equipment transferred to Dakar, Senegal after station phase out after STS-2
Santiago, Chile	33°09'S 70°40'W	1957	1988	Minitrack, MOTS, SATAN, GRARR, USB, a 30 and 40-foot dish	Most extensive of the South American stations. Operated jointly with the University of Chile who continues to run the station today.
Europe					
Winkfield, England	57°27'N 00°42'E	1961	1981	Minitrack, 14-foot dish, SATAN, MOTS	Operated by the British under agreement to the U.S.
Africa					
Johannesburg, South Africa	25°53'S 27°42'E	1957	1975	Minitrack, MOTS, SATAN, a 40 and 85-foot dish	Operated by South African nationals. Located with DSN site. Phased out under political pressure.
Majunga/ Tananarive, Malagasy Republic	19°S 47°18'E	1964	1975	45-foot dish, GRARR	Transportable site erected at Majunga which was moved a year later to a more permanent establishment at Tananarive to support the MSFN. Forced out due to political strife.
Australia					
Carnarvon, Western Australia	24°54'S 113°43'E	1964	1974	GRARR, C-band radar, 30-foot USB	Co-located with a MSFN site.
Darwin, Northern Territory	12°17'S 130°49'E	1965	1969	14-foot antenna to support OGO project	Temporary mobile OGO project station. Antenna transferred to Kauai MSFN station.
Orroral Valley, Australian Capital Territory	35°38'S 148°57'E	1965	1984	Minitrack, MOTS, two SATANs, 85-foot dish	The 85-foot antenna was relocated to the University of Tasmania after the station closed where it is still in use by the school
Toowoomba/ Cooby Creek, Queensland	27°24'S 151°56'E	1966	1969	SATAN, ATS	Temporary site used to support the ATS project
Woomera, South Australia	31°23'S 136°53'E	1957	1966	Minitrack, MOTS, SATAN	Closed out within 12 months after Orroral Valley became operational. Minitrack equipment transferred to Orroral.

Table 2-1 is a summary of the STADAN ground stations highlighting their capabilities and their roles in the network. Appendix 1 shows the network as it appeared throughout the 1960s into the mid-1970s.

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Network stations were operated by either American contractors, local nationals (if overseas), or a combination of both. To promote international goodwill, NASA used local nationals to the extent possible for non-technical positions. The staff usually worked under the direction of NASA civil service supervisors assigned to the station, led by a Station Director and sometimes a second-in-charge, the Deputy Station Director.[51] While Goddard contracted with several companies to perform station support requirements that ranged from facility maintenance to transportation to administration, it generally relied on two commercial contractors to run the network: BFEC and RCA.[52]

As the tracking network expanded from Minitrack to STADAN, so did the level of contractor support provided. In FY1963, Bendix as a corporation did $32.5 million of business with NASA. Over the next five years, as the STADAN expanded and became more complex, more stations were added. By FY 1968, Bendix was doing $123.8 million worth of work for NASA, making it the space agency's seventh top contractor in overall dollar amount obligated.[53] Of this amount, $39.1 million, or about one-third, were two prime contracts with Goddard (NAS5-9968 and NAS5-10750) to operate and maintain its two networks, the STADAN and the MSFN.[54]

Bendix employees received a number of incentives to work at overseas locations. By the late 1960s, for example, an array of seven allowances, each tailored to the specific conditions of the locale, was being offered. One, the Foreign Assignment Allowance, was a 10 percent of base pay "sweetener" applied to compensate for the inconveniences and cultural disadvantages at locations where employees had to learn a foreign language and assume additional assignments due to lack of trained local personnel. The Cost of Living Allowance was granted in circumstances where costs of subsistence, services and goods exceeded comparable expenses in the Baltimore-Washington area. Since that particular variable allowance did not cover housing expenses, a separate Housing Allowance was also established. For contractor personnel on location with families, many companies provided an Education Allowance that reimbursed the cost of providing educational services to children at local schools who would otherwise be tax-subsidized in United States public schools. As the years went by, many contractor employees ended up spending their whole careers overseas, doing very well for themselves, moving from station to station as the landscape of the tracking network evolved.[55]

A hallmark of STADAN was the training that its people received. Goddard and the contractor teams hired both degreed and nondegreed personnel in various disciplines, although most were in the field of electronics and electrical engineering. Technical training had been taking place at Blossom Point since the inception of Minitrack in 1956. In the fall of 1966, as NASA ceased joint satellite tracking operations with the Navy at Blossom Point, satellite training operations were consolidated with manned network training at Wallops Island. This was subsequently relocated to the Network Test and Training Facility (NTTF) at the GSFC itself.[56]

The NTTF employed about 40 contractor instructors who held either a Bachelor of Science degree or a teaching certificate in the appropriate technical field. The program was modeled along the lines of a vocational school, with a format for both classroom instruction and hands-on laboratory training. The instructional emphasis was decidedly pragmatic, fostering in students the capacity to quickly adjust and adapt to changing situations which might arise in the field. Instructors were sometimes dispatched to field stations themselves, providing the opportunity to conduct on-site training with foreign local nationals.

To ensure quality control and technical competency, Goddard, in the fall of 1967, instituted a formal job certification program for American citizens assigned to field stations throughout the network. All individuals studying at the NTTF were awarded primary certification in one technical area prior to his or her graduation. Certification was based on both academics and hands-on test performance in technical skill areas such as telemetry, communications, teletype operations, the IBM 1218 computer, and the FPQ-6 radar just to name a few. Certification tests examined the readiness of a technician in four fundamental areas: 1) Safety Procedures, 2) Operating Procedures, 3) Preventive Maintenance, and 4) Corrective Maintenance. Incumbent personnel (those veterans already serving on-site prior to the creation of the Certification Program) in some cases received certification by waiver based on their field experience; otherwise, they too had to take the test. Primary certification in a particular specialty meant that a person could practice that skill at any network location. The program also allowed personnel to obtain "secondary certification." This feature allowed station staff members to receive a number of secondary certifications in different technical skill areas outside their primary area. But these were generally limited for use only at the particular station where they worked.

The program was very successful. So successful, in fact, that three years after it was implemented, in a move to further promote foreign cooperation through the hiring of local labor, the Certification Program was extended to include skilled positions. Initial implementation began in 1970 at

Chile, Ecuador, and Madagascar. It eventually resulted in more than 200 local nationals being certified, mostly for Electronics Technician positions.[57]

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Ground stations and the people who operate them, however, do not make a network by themselves. A network requires a coherent and integrated communication system that enables the stations to talk to each other and to the GSFC. The NASA Communications Network, known as NASCOM, was that critical ground link which tied the system together. But when NASA first constructed its networks, NASCOM did not yet exist. Initially, each of the Agency's three networks—the Satellite Tracking And Data Acquisition Network (STADAN), the Manned Space Flight Network (MSFN) and the Deep Space Network (DSN)—all had their own communication system linking the ground stations to the control center. Even at Goddard, the satellite and human spaceflight communication organizations were in separate divisions within John Mengel's Tracking and Data Systems Directorate.

But by 1962, complexity of both the STADAN and the DSN had noticeably increased. With the new MSFN coming online, it became clear that running three separate communication systems at the same time was just not going to be very efficient, both technically and in cost. Technically, economically and logistically, a single system made much more sense. Studies were done to this effect which showed significant savings could in fact be achieved through circuit sharing and common use of field equipment. In July 1963, the Communications Division was formed at Goddard under the direction of Lavern R. "Vern" Stelter—who later became Chief of the NASA Communications Division—to coordinate all Agencies' ground communication activities. Up until then, this had been done by Headquarters.

In describing how the tracking networks worked, some have likened it to the human body. That is, if the control centers were the brain, then the tracking stations were the eyes and ears and the NASCOM lines the central nervous system. The flow of information between the worldwide network of stations and the operations center at the GSFC during a mission was quite extensive. The degree of mission success often hinged on the ability of a station to transmit critical information quickly and accurately. The NASCOM voice circuits enabled Goddard to manage, exercise, coordinate and brief all stations simultaneously during simulations and on actual missions. It also enabled the stations to discuss problems and procedures among themselves.

Even before the formation of NASA, John Mengel's Minitrack team at the NRL was faced with the job of getting tracking data back from the picket line and the other stations around the world. To develop and operate this system, Mengel recruited experienced range workers from Edwards

(known as Muroc at the time) in Southern California, White Sands in Southern New Mexico, and from industry. As personnel from the military ranges and surveillance networks brought their knowledge to the NRL, its pool of communications knowledge quickly grew. This was vital since many of the then proposed Minitrack stations had to be in either undeveloped parts of the world or at great distances like Australia.

At first, the challenge was just getting data back to NRL. Speed was not too important. Similarly, science data was not time critical and were recorded on 14-track magnetic tapes and airmailed back to the United States. Within the continental United States, existing teletype lines were found to support Minitrack stations at San Diego, Blossom Point, and Fort Stewart, all of which were on established military installations. Not surprisingly, the real communication challenges were in South America where lines had to be added in uninhabited rainforests and undeveloped terrain. Throughout the 1940s and 1950s, the United States had worked with Latin American countries as part of its Inter-American Geodetic Survey (IAGS) to produce comprehensive maps of the entire Western Hemisphere. The U.S. government was able to cultivate important relationships through this endeavor which enabled the Army to go into some of these countries and lay communication lines that were needed by the South American stations. Stelter explained how it all worked out:

> Aside from the tremendous technical advances which made this network possible, we had the unqualified support and cooperation of every foreign country involved. Without their support, there would have been no successful NASA Communications Network. Indeed, if we had to pay for every person involved in this worldwide effort, we could not have afforded it. At that, our phone bill during the Apollo program was $50 million per year! On our part, we tried our utmost to make our foreign colleagues full participants and they in turn provided hundreds, possibly thousands of people, to assure the reliability and performance of the NASCOM network on which our success, and often the lives of our astronauts, depended.[58]

All of NASA—not just STADAN and the other tracking networks—relied on the NASCOM for communications, everything from everyday telephone conference calls to high rate telemetry transmissions. To this end, a combination of permanently rented circuits for teletype and voice were used. These included commercial landlines operated by telephone companies, ocean-floor cables, high frequency radio links and microwave relays,

and finally, NASA and commercial communications satellites. By 1969, some two million miles of circuits in one form or another were being used.[59]

Data was routed using these circuits to key switching centers at various hubs in different parts of the world—Canberra, London, Madrid, Pasadena—which piped the data back to the United States.[60] Said Stelter:

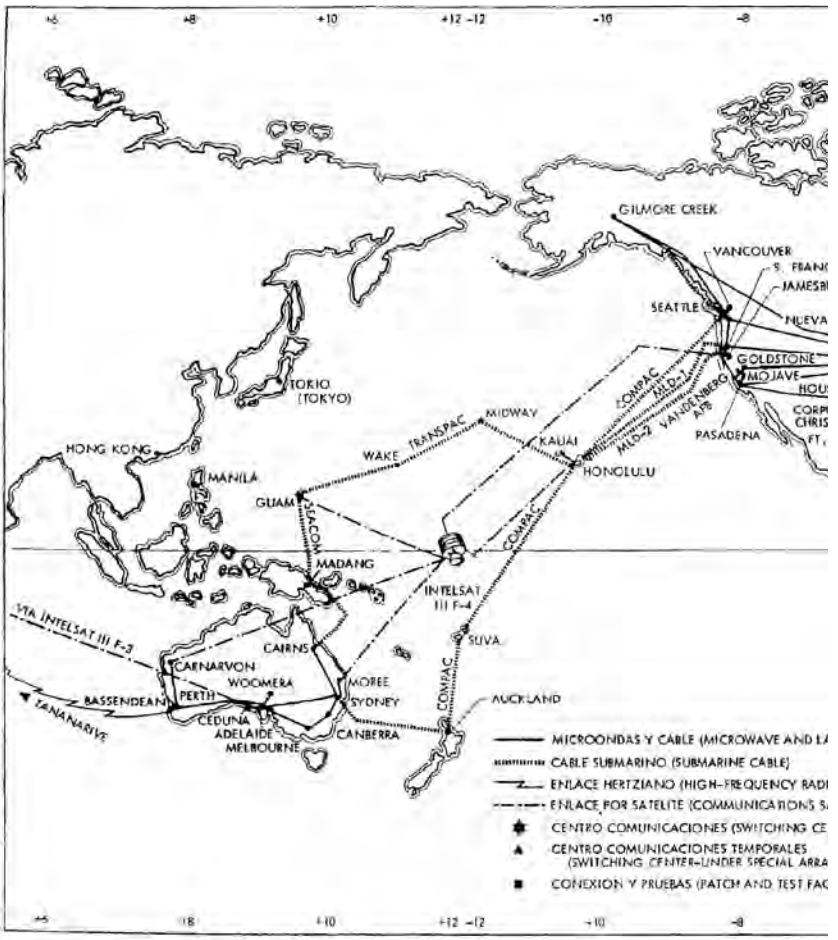
> Initially, we had to rely on high frequency radio and on other rather primitive transmission technology which was anything but reliable. Somehow, we were able to make it work. . . . Later, we added the first computerized switching facilities, developed by UNIVAC [UNIVersal Automatic Computer], which was a significant advance. It assured us that all data from the spacecraft would get to the user in a reliable manner and outbound commands would in turn reach the spacecraft quickly. No easy task![61]

Other communication systems existed, but NASCOM was by far the world's largest, real-time communications system. It was very much a prototype, pioneering the broad-band communication systems of today.

As Minitrack steadily evolved into STADAN in the 1960s, the real-time link between satellites and controllers back at Goddard also improved. This was still at a time when very few commandable features were actually built into satellites. While things like tape recorders could be commanded to read out the data they had stored, little else could be directed from the ground. With the introduction of more complex satellites, particularly the OGO and OAO, by the mid-1960s mission controllers possessed the command capability for controlling many more satellite functions in addition to just the onboard scientific experiments. Goddard controllers, linked directly to STADAN stations, could now in effect "drive" a satellite via commands uplinked through a station as a satellite passed overhead.[62]

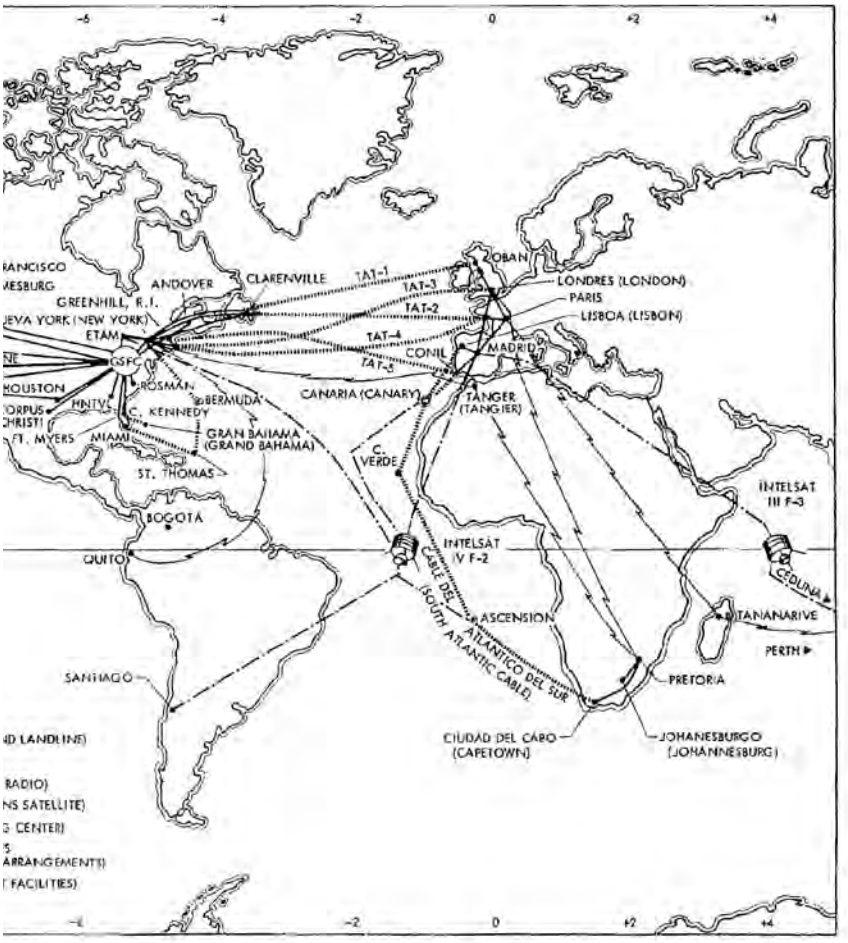
For Minitrack, transmission speed was a nicety but not a requirement. Now, with the advent of this new class of large observatory satellites and their high-rate data dumps, speed became not only desired but necessary. In addition, with Project Mercury on the horizon calling for real-time voice and tracking, telemetry, and command capabilities, a change was needed to bring the old NASCOM links literally "up to speed."

Preparations for the manned-satellite program, in particular, drove several requirements. It needed real-time, wide-band communications—the first for quick trajectory computations to support mission decisions and the second for the heavy data volume required to monitor spacecraft health and status and to talk with the astronaut. A most obvious requirement was that ground-based flight controllers, some of whom were stationed at tracking sites, had to be able to talk directly with the astronaut during a pass. Voice



If tracking stations were the sensory organs of NASA's communications network and the Goddard Space Flight Center the brain, then NASCOM (the NASA Communications Network) could certainly be called its central nervous system, with over 2 million circuit

circuits between the spacecraft and the ground, therefore, had to be added to the system for the first time. (Television coverage inside the Mercury capsule was also proposed but would not be implemented until Apollo, partly because the communication technology was just not yet ready to handle the demands of such a bandwidth intensive item.) Added to all this was perhaps the most important difference between satellite and human spaceflight communications: reliability. If communications were lost during a pass for an observatory satellite, some data might be lost; on a human flight, the consequences could be much more severe.[63]



miles at its height of operations. Shown here is a layout of the circuits in 1972. Since the diagram was taken from a brochure describing the Madrid Station, some of the notes are in Spanish. (Madrid Space Station, INTA/NASA: Information Brochure, JPL P72-223)

These important differences were known to Langley engineers even in 1958 as they laid the foundation for a Mercury Network. It soon became apparent, however, that they could not just simply piece together a world-wide, real-time communication system from existing commercial or military circuits. Integration with Minitrack was also not possible; the requirements were just too different. Communication requirements for Mercury were, at the time, such a leap from anything else that had been done that Langley, and later Goddard, knew that an entirely different network had to be designed and built specifically to support the new man-in-space project.

This network, as it finally emerged, would consist of two communication circuits, one for voice and one for teletype. Teletype would use the then state-of-the-art Western Union 111 Torn-Tape Relay System, limited to a speed of 60 words per minute. The limitation was due to capabilities of the terminal equipment that was already being installed at the new MSFN sites. Initially, oversea switching centers were located at Adelaide, Australia (subsequently moved to Canberra), Honolulu, and London. Later, communication switching centers were added in Madrid, Guam, and the Kennedy Space Center. Goddard served as the hub and main switching center.[64]

"Data pipe" between Goddard and the Cape represented a substantial advance in wideband transmission capability over the previous generation with NASCOM. In 1961, for instance, four two-way, voice data circuits each capable of transmitting at one kilobit-per-second (1,000 bps) were installed between the Cape and Goddard. While this may not seem like something hardly even worth mentioning in today's world where data rates are measured in *gigabits*-per-second (that is, a billion bps), it was quite the improvement over the Minitrack days; one of these channels could now transmit 30 times as much data. This faster data requirement was needed because, prior to the establishment of the Mission Control Center in Houston, Goddard had to control the real-time displays at Mercury Control, then still located at Cape Canaveral. Soon, other wideband data lines were installed between GSFC and the STADAN stations, most notably at the new DAF of Rosman and Alaska. Finally, in preparation for sending astronauts into space, another important voice link was established in 1963 with a newly laid ocean floor cable connecting the new station on Bermuda with the Cape. This link could carry 2,000 bits-per-second of digital information and would continue to serve the Bermuda Station well into the Shuttle era.[65]

★ ★ ★

As NASA began sending men to the Moon, new stations were added and aircraft and ships were assigned to the network. Unlike Mercury or Gemini, live television was going to be used on Apollo and new equipment accommodating state-of-the-art transmission schemes at the S-band frequencies would be installed throughout the network. The role of communication satellites like *Syncom* would be called on to provide "switchboards in the sky." Developed by an industry team, their success would lead to the creation of the Cosmsat Corporation and an international consortium to manage and market worldwide satellite communication services. In time, this development would pay great dividends as the consortium's Intelsat ultimately ended up serving the very government agencies that helped to develop its technology.[66]

To see how sending humans into space remade the Goddard networks, one needs to turn the clock back to a time before there was a GSFC.